

Brutal Practice Paper B32 - Winds, Storms & Cyclones | Comparing Quantities | Soil | Exponents & Powers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A coastal weather station clocks a cyclone's winds at 108 kilometres per hour. Expressed in metres per second, that wind speed is? [\[Winds, Storms & Cyclones\]](#)

- (A) 3 m/s
- (B) 300 m/s
- (C) 1.8 m/s
- (D) 30 m/s
- (E) 36 m/s
- (F) 60 m/s
- (G) 10.8 m/s

2. A shopkeeper buys a fan for Rs 800 and wants a profit of 25 percent. The selling price he must mark is?

[\[Comparing Quantities\]](#)

- (A) Rs 825
- (B) Rs 600
- (C) Rs 200
- (D) Rs 1200
- (E) Rs 1100
- (F) Rs 1000
- (G) Rs 1025

3. In a percolation test, 200 millilitres of water sinks through a soil sample in 25 minutes. The percolation rate of the soil is? [\[Soil\]](#)

- (A) 8 min/mL
- (B) 225 mL/min
- (C) 16 mL/min
- (D) 0.125 mL/min
- (E) 8 mL/min
- (F) 4 mL/min

4. Using the law for multiplying powers with the same base, the value of 2 cubed multiplied by 2 to the fourth power is? [\[Exponents & Powers\]](#)

- (A) 128
- (B) 16384
- (C) 2187
- (D) 256
- (E) 127
- (F) 32
- (G) 64

5. Warm air over a sun-baked field rises because heating makes air expand. If 40 cubic metres of this air expands by 15 percent, its new volume is? [\[Winds, Storms & Cyclones\]](#)

- (A) 46 m³
- (B) 34 m³
- (C) 40.15 m³
- (D) 43 m³
- (E) 54 m³
- (F) 6 m³
- (G) 60 m³

6. A jacket is marked at Rs 1200 and offered with a 20 percent discount. The price a customer actually pays is? [\[Comparing Quantities\]](#)

- (A) Rs 1180
- (B) Rs 1440
- (C) Rs 1000
- (D) Rs 1220
- (E) Rs 980
- (F) Rs 240
- (G) Rs 960

7. A dry soil sample of mass 250 grams can hold 100 grams of water before dripping. Its water-holding capacity, as a percentage of the dry mass, is? [\[Soil\]](#)

- (A) 60%
- (B) 2.5%
- (C) 45%
- (D) 25%
- (E) 140%
- (F) 100%
- (G) 40%

8. Applying the power-of-a-power rule, the value of 3 squared, all raised to the power 3, is? [\[Exponents & Powers\]](#)

- (A) 27
- (B) 81
- (C) 243
- (D) 2187
- (E) 729
- (F) 216
- (G) 18

9. Normal sea-level air pressure is 1008 hectopascals, but at a storm's centre it falls to 968 hectopascals. The drop in pressure is? [\[Winds, Storms & Cyclones\]](#)

- (A) 1976 hPa
- (B) 4 hPa
- (C) 48 hPa
- (D) 40 hPa
- (E) 60 hPa
- (F) 32 hPa
- (G) 140 hPa

10. A coat marked at Rs 2000 is sold with two successive discounts of 10 percent and then 5 percent. The final price is? [\[Comparing Quantities\]](#)

- (A) Rs 1650
- (B) Rs 290
- (C) Rs 1800
- (D) Rs 1700
- (E) Rs 1900
- (F) Rs 1500
- (G) Rs 1710

11. A 500 gram sample of garden soil contains 25 grams of humus. The percentage of humus in the soil is? [\[Soil\]](#)

- (A) 2.5%
- (B) 50%
- (C) 5%
- (D) 525 grams
- (E) 0.5%
- (F) 20%

12. Dividing powers with the same base, the value of 5 to the seventh power divided by 5 to the fourth power is? [\[Exponents & Powers\]](#)

- (A) 25
- (B) 125
- (C) 15625
- (D) 625
- (E) 78125
- (F) 1000

13. Normal pressure is 1000 hectopascals. Inside a deep cyclone it sinks to 920 hectopascals. By what percentage has the pressure fallen? [\[Winds, Storms & Cyclones\]](#)

- (A) 92%
- (B) 920%
- (C) 20%
- (D) 8%
- (E) 0.8%
- (F) 12%

14. An item costing Rs 400 is sold for Rs 500. The profit percentage on the cost is? [\[Comparing Quantities\]](#)

- (A) 40%
- (B) 100%
- (C) 50%
- (D) 25%
- (E) 125%
- (F) 10%

15. A soil profile has a topsoil layer 15 centimetres deep above a subsoil layer 45 centimetres deep. The combined depth of these two layers is? [\[Soil\]](#)

- (A) 50 cm
- (B) 60 cm
- (C) 75 cm
- (D) 675 cm
- (E) 3 cm
- (F) 6 m

16. Written in standard (scientific) form, the number 56000 is? [\[Exponents & Powers\]](#)

- (A) 56×10^3
- (B) 5.6×10^2
- (C) 5.6×10^5
- (D) 56×10^4
- (E) 5.6×10^4
- (F) 0.56×10^5
- (G) 5.6×10^3

17. A cyclone's centre travels 360 kilometres across the sea in 8 hours. Its average speed of advance is?

[\[Winds, Storms & Cyclones\]](#)

- (A) 50 km/h
- (B) 40 km/h
- (C) 45 km/h
- (D) 52 km/h
- (E) 368 km/h
- (F) 36 km/h

18. Rs 5000 is lent at a simple interest rate of 8 percent per year for 3 years. The interest earned is?

[\[Comparing Quantities\]](#)

- (A) Rs 1200
- (B) Rs 800
- (C) Rs 13000
- (D) Rs 120
- (E) Rs 6200
- (F) Rs 400

19. Sandy soil drains 200 millilitres in 10 minutes while clay soil drains the same amount in 40 minutes.

The ratio of the sandy soil's percolation rate to the clay soil's is? [\[Soil\]](#)

- (A) 10 : 40
- (B) 1 : 4
- (C) 4 : 1
- (D) 2 : 1
- (E) 1 : 2
- (F) 5 : 1
- (G) 20 : 5

20. The value of 2 raised to the tenth power is? [\[Exponents & Powers\]](#)

- (A) 1000
- (B) 512
- (C) 256
- (D) 20
- (E) 1024
- (F) 100
- (G) 2048

21. A storm is advancing at 45 kilometres per hour. What is this speed in metres per second? [\[Winds, Storms & Cyclones\]](#)

- (A) 125 m/s
- (B) 7.5 m/s
- (C) 16.2 m/s
- (D) 1.25 m/s
- (E) 45 m/s
- (F) 162 m/s
- (G) 12.5 m/s

22. Rs 5000 is invested at 8 percent simple interest per year for 3 years. The total amount returned at the end is? [\[Comparing Quantities\]](#)

- (A) Rs 6800
- (B) Rs 6200
- (C) Rs 1200
- (D) Rs 13000
- (E) Rs 6000
- (F) Rs 5240

23. A wet soil sample weighs 300 grams; after drying it weighs 240 grams. The percentage of its original mass that was water is? [\[Soil\]](#)

- (A) 24%
- (B) 12%
- (C) 25%
- (D) 80%
- (E) 60%
- (F) 30%
- (G) 20%

24. Using the rule that a product raised to a power spreads over each factor, the value of (2×3) all cubed is? [\[Exponents & Powers\]](#)

- (A) 144
- (B) 27
- (C) 18
- (D) 72
- (E) 36
- (F) 216

- 25.** A cyclone 600 kilometres offshore is moving toward the coast at a steady 25 kilometres per hour. After how many hours will it reach land? [\[Winds, Storms & Cyclones\]](#)
- (A) 15 hours
 - (B) 48 hours
 - (C) 30 hours
 - (D) 24 hours
 - (E) 2.4 hours
 - (F) 625 hours
- 26.** In a class the ratio of boys to girls is 3 : 5. What percentage of the whole class are boys? [\[Comparing Quantities\]](#)
- (A) 60%
 - (B) 53%
 - (C) 37.5%
 - (D) 62.5%
 - (E) 40%
 - (F) 30%
 - (G) 35%
- 27.** One soil sample has pH 5 and another has pH 8. On the pH scale, the difference between the two readings is? [\[Soil\]](#)
- (A) 4 units
 - (B) 40 units
 - (C) 5 units
 - (D) 2 units
 - (E) 1.6 units
 - (F) 3 units
 - (G) 13 units
- 28.** Both 7 to the power 0 and 3 to the power 0 are added together. Their sum is? [\[Exponents & Powers\]](#)
- (A) 3
 - (B) 2
 - (C) 1
 - (D) 49
 - (E) 0
 - (F) 21
 - (G) 10
- 29.** A heated parcel of air has its density drop from 1.2 to 0.9 kilograms per cubic metre as it warms. The percentage decrease in density is? [\[Winds, Storms & Cyclones\]](#)
- (A) 25%
 - (B) 75%
 - (C) 30%
 - (D) 0.3%
 - (E) 20%
 - (F) 33.3%

30. An employee's monthly salary of Rs 20000 is raised by 15 percent. The new monthly salary is?

[Comparing Quantities]

- (A) Rs 22000
- (B) Rs 20015
- (C) Rs 23150
- (D) Rs 17000
- (E) Rs 3000
- (F) Rs 23000
- (G) Rs 30000

31. A farmer spreads compost at 2 kilograms per square metre over a field of 50 square metres. The total mass of compost needed is? [Soil]

- (A) 2500 kg
- (B) 100 kg
- (C) 200 kg
- (D) 102 kg
- (E) 52 kg
- (F) 1000 kg

32. Negative bases follow the sign rule for odd powers. The value of negative 2 raised to the fifth power is?

[Exponents & Powers]

- (A) -16
- (B) 32
- (C) -25
- (D) -10
- (E) -32
- (F) 10
- (G) -31

33. The monsoon delivers 80 centimetres of rain spread evenly over 4 months. The average rainfall per month is? [Winds, Storms & Cyclones]

- (A) 320 cm
- (B) 10 cm
- (C) 25 cm
- (D) 20 cm
- (E) 16 cm
- (F) 40 cm
- (G) 84 cm

34. A phone is priced at Rs 12000 before an 18 percent tax is added at the counter. The amount the buyer pays is? [Comparing Quantities]

- (A) Rs 14180
- (B) Rs 9840
- (C) Rs 2160
- (D) Rs 13160
- (E) Rs 14000
- (F) Rs 14160

- 35.** Weathering wears down a rock surface by about 2 millimetres each year. Over 50 years, the depth of rock worn away is? [\[Soil\]](#)
- (A) 100 mm
 - (B) 10 mm
 - (C) 48 mm
 - (D) 20 mm
 - (E) 1000 mm
 - (F) 52 mm
 - (G) 25 mm
- 36.** Simplifying with exponent laws, the value of 2 to the eighth power divided by the product of 2 cubed and 2 squared is? [\[Exponents & Powers\]](#)
- (A) 32
 - (B) 16
 - (C) 1
 - (D) 4
 - (E) 2
 - (F) 8
- 37.** During a downpour, 20 centimetres of rain falls evenly on each square metre of ground. How many litres of water land on one square metre? [\[Winds, Storms & Cyclones\]](#)
- (A) 100 L
 - (B) 20000 L
 - (C) 200 L
 - (D) 0.2 L
 - (E) 2 L
 - (F) 2000 L
- 38.** A toy is sold for Rs 1100 at a profit of 10 percent. The cost price of the toy was? [\[Comparing Quantities\]](#)
- (A) Rs 990
 - (B) Rs 1000
 - (C) Rs 1100
 - (D) Rs 1210
 - (E) Rs 100
 - (F) Rs 900
 - (G) Rs 1010
- 39.** A loam soil is 60 percent sand, 30 percent silt and 10 percent clay by mass. In a 400 gram sample, the mass of clay is? [\[Soil\]](#)
- (A) 40 g
 - (B) 10 g
 - (C) 120 g
 - (D) 4 g
 - (E) 240 g
 - (F) 60 g
 - (G) 100 g

40. Expressed as a single power of 2, the number 1024 is? [\[Exponents & Powers\]](#)

- (A) 2^{11}
- (B) 2^8
- (C) 2^9
- (D) 2^5
- (E) 4^{10}
- (F) 10^2
- (G) 2^{10}

41. A storm surge pushes the sea 3.5 metres above its normal high-tide level of 1.8 metres. The total water height reached is? [\[Winds, Storms & Cyclones\]](#)

- (A) 1.7 m
- (B) 4.3 m
- (C) 3.5 m
- (D) 2.3 m
- (E) 53 m
- (F) 5.3 m
- (G) 6.3 m

42. A bag is marked at Rs 750 but sold for Rs 600. The discount percentage given is? [\[Comparing Quantities\]](#)

- (A) 18%
- (B) 30%
- (C) 150%
- (D) 25%
- (E) 12.5%
- (F) 15%
- (G) 20%

43. Rainwater soaks into a field at a steady 18 centimetres in 6 hours. The infiltration rate is? [\[Soil\]](#)

- (A) 0.33 cm/h
- (B) 3 cm/h
- (C) 24 cm/h
- (D) 108 cm/h
- (E) 12 cm/h
- (F) 9 cm/h

44. Rewriting with a common base, 4 cubed equals 2 to which power, and what is its value? [\[Exponents & Powers\]](#)

- (A) $2^8 = 256$
- (B) $4^6 = 4096$
- (C) $2^5 = 32$
- (D) $2^{12} = 4096$
- (E) $2^3 = 8$
- (F) $2^6 = 64$
- (G) $2^7 = 128$

45. Two cyclones have peak winds of 120 and 80 kilometres per hour. The ratio of the faster wind speed to the slower, in simplest form, is? [\[Winds, Storms & Cyclones\]](#)
- (A) 5 : 3
 - (B) 3 : 2
 - (C) 2 : 3
 - (D) 120 : 80
 - (E) 4 : 3
 - (F) 3 : 1
 - (G) 6 : 4
46. Rs 4500 is shared among three friends in the ratio 2 : 3 : 4. The largest share is? [\[Comparing Quantities\]](#)
- (A) Rs 2500
 - (B) Rs 2000
 - (C) Rs 2250
 - (D) Rs 500
 - (E) Rs 1800
 - (F) Rs 1000
 - (G) Rs 1500
47. A soil mixture has sand and clay in the ratio 7 : 3. In a 500 gram batch, the mass of sand is? [\[Soil\]](#)
- (A) 357 g
 - (B) 210 g
 - (C) 300 g
 - (D) 250 g
 - (E) 350 g
 - (F) 150 g
 - (G) 70 g
48. Written in standard form, the large number 34000000 is? [\[Exponents & Powers\]](#)
- (A) 3.4×10^7
 - (B) 3.4×10^6
 - (C) 3.4×10^5
 - (D) 3.4×10^8
 - (E) 0.34×10^8
 - (F) 34×10^7
 - (G) 34×10^6
49. An anemometer's cups spin around 150 times every minute in a steady wind. How many turns will they complete in 10 seconds? [\[Winds, Storms & Cyclones\]](#)
- (A) 15 turns
 - (B) 1500 turns
 - (C) 25 turns
 - (D) 90 turns
 - (E) 250 turns
 - (F) 9 turns
 - (G) 2500 turns

50. A shop first takes 20 percent of 50 percent of a Rs 800 bill as a special rebate. The rebate amount is?

[Comparing Quantities]

- (A) Rs 100
- (B) Rs 160
- (C) Rs 80
- (D) Rs 560
- (E) Rs 720
- (F) Rs 40
- (G) Rs 400

51. A field of 1 hectare is to receive fertiliser at 5 grams per square metre. Given that 1 hectare is 10000 square metres, the total fertiliser needed is? [Soil]

- (A) 50000 kg
- (B) 2000 kg
- (C) 5 kg
- (D) 500 kg
- (E) 50 g
- (F) 50 kg
- (G) 10 kg

52. Comparing the two powers 2 to the fifth and 5 to the second, which is larger and by how much?

[Exponents & Powers]

- (A) 5^2 is larger by 3
- (B) 2^5 is larger by 7
- (C) they are equal
- (D) 5^2 is larger by 7
- (E) 2^5 is larger by 57
- (F) 2^5 is larger by 25

53. Wind blows because a pressure difference pushes on the air. If a pressure difference of 150 pascals acts on a window of area 2 square metres, the force is? [Winds, Storms & Cyclones]

- (A) 3000 N
- (B) 300 N
- (C) 600 N
- (D) 30 N
- (E) 148 N
- (F) 75 N
- (G) 450 N

54. A town's population of 8000 grows by 5 percent in the first year and a further 5 percent in the second year. Its population after two years is? [Comparing Quantities]

- (A) 8400
- (B) 8820
- (C) 8410
- (D) 9000
- (E) 820
- (F) 8800
- (G) 8000

55. Sea water seeping into a field leaves 4 grams of salt in every 1000 grams of soil. The salt content as a percentage is? [\[Soil\]](#)

- (A) 250%
- (B) 2.5%
- (C) 4%
- (D) 0.4%
- (E) 0.04%
- (F) 40%
- (G) 1%

56. A fraction raised to a power applies to top and bottom alike. The value of one-half cubed is? [\[Exponents & Powers\]](#)

- (A) $\frac{1}{9}$
- (B) $\frac{2}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{3}{8}$
- (E) $\frac{3}{2}$
- (F) $\frac{1}{6}$
- (G) $\frac{1}{8}$

57. On a hot afternoon a sea breeze carries cool air 15 kilometres inland in 30 minutes. Its speed is? [\[Winds, Storms & Cyclones\]](#)

- (A) 15 km/h
- (B) 30 km/h
- (C) 60 km/h
- (D) 7.5 km/h
- (E) 0.5 km/h
- (F) 450 km/h
- (G) 45 km/h

58. A trader buys a chair for Rs 600 and sells it for Rs 480. The loss percentage is? [\[Comparing Quantities\]](#)

- (A) 24%
- (B) 25%
- (C) 12%
- (D) 20%
- (E) 120%
- (F) 15%
- (G) 30%

59. A drying soil sample loses 30 grams of water out of an original 150 grams of water it held. The percentage of its water that has evaporated is? [\[Soil\]](#)

- (A) 30%
- (B) 15%
- (C) 20%
- (D) 5%
- (E) 80%
- (F) 25%

60. Using exponent laws on a common base, 9 squared multiplied by 9 equals which single power of 3, and its value is? [\[Exponents & Powers\]](#)

- (A) $3^4 = 81$
- (B) $3^8 = 6561$
- (C) $9^2 = 81$
- (D) $9^3 = 729$
- (E) $3^5 = 243$
- (F) $3^3 = 27$
- (G) $3^6 = 729$

61. As a storm strengthens, its wind speed climbs from 60 to 90 kilometres per hour. The percentage increase in wind speed is? [\[Winds, Storms & Cyclones\]](#)

- (A) 60%
- (B) 50%
- (C) 45%
- (D) 25%
- (E) 30%
- (F) 33.3%
- (G) 150%

62. On a deposit of Rs 2000, the simple interest earned over 3 years is Rs 480. The yearly interest rate is? [\[Comparing Quantities\]](#)

- (A) 8%
- (B) 24%
- (C) 6%
- (D) 4%
- (E) 10%
- (F) 16%
- (G) 12%

63. A pit dug 1.2 metres deep reaches the water table at 80 centimetres. How much further, in centimetres, must it be dug to reach the bottom of the pit at 1.2 metres? [\[Soil\]](#)

- (A) 32 cm
- (B) 20 cm
- (C) 40 cm
- (D) 0.4 cm
- (E) 48 cm
- (F) 4 cm
- (G) 200 cm

64. Dividing powers of the same base, 6 to the fourth power divided by 6 squared equals? [\[Exponents & Powers\]](#)

- (A) 36
- (B) 12
- (C) 46656
- (D) 18
- (E) 216
- (F) 1296
- (G) 6

65. A rain gauge collects 45 millimetres of rain over 3 hours of steady rainfall. The rainfall rate is? [\[Winds, Storms & Cyclones\]](#)

- (A) 5 mm/h
- (B) 30 mm/h
- (C) 135 mm/h
- (D) 9 mm/h
- (E) 48 mm/h
- (F) 15 mm/h

66. A sum of Rs 1500 earns Rs 450 as simple interest at 10 percent per year. For how many years was it invested? [\[Comparing Quantities\]](#)

- (A) 2 years
- (B) 30 years
- (C) 6 years
- (D) 1 year
- (E) 4.5 years
- (F) 3 years
- (G) 5 years

67. Of the volume of a loose soil, about 45 percent is empty pore space. In a 2000 cubic centimetre sample, the volume of pore space is? [\[Soil\]](#)

- (A) 2045 cm³
- (B) 900 cm³
- (C) 1100 cm³
- (D) 45 cm³
- (E) 1000 cm³
- (F) 90 cm³
- (G) 450 cm³

68. Multiplying powers with the same exponent, 2 cubed times 5 cubed can be written as a single power of 10 with what value? [\[Exponents & Powers\]](#)

- (A) 125
- (B) 1000
- (C) 30
- (D) 15
- (E) 10000
- (F) 216
- (G) 100

69. If a heavy storm keeps raining at 15 millimetres per hour for 18 hours, the total rainfall is? [\[Winds, Storms & Cyclones\]](#)

- (A) 27 mm
- (B) 135 mm
- (C) 33 mm
- (D) 1.2 mm
- (E) 270 mm
- (F) 240 mm
- (G) 2700 mm

70. An item costing Rs 1000 is marked up by 50 percent, then sold at a 20 percent discount off that marked price. The overall profit percentage is? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 15%
- (C) 10%
- (D) 50%
- (E) 40%
- (F) 20%
- (G) 30%

71. To make a planting mix, soil and compost are blended in the ratio 5 : 1 by mass. To prepare 18 kilograms of mix, the mass of compost required is? [\[Soil\]](#)

- (A) 6 kg
- (B) 9 kg
- (C) 1 kg
- (D) 2 kg
- (E) 3.6 kg
- (F) 3 kg

72. Applying the power-of-a-power rule, 10 cubed all squared equals which power of 10, and its value is? [\[Exponents & Powers\]](#)

- (A) $10^6 = 1000000$
- (B) $10^8 = 100000000$
- (C) $10^7 = 10000000$
- (D) $10^6 = 100000$
- (E) $10^9 = \text{a billion}$
- (F) $10^5 = 1000000$
- (G) $10^5 = 100000$

73. A town of 20000 people must be evacuated before a cyclone using buses that each carry 50 people. The minimum number of buses needed is? [\[Winds, Storms & Cyclones\]](#)

- (A) 100 buses
- (B) 1000050 buses
- (C) 40 buses
- (D) 400 buses
- (E) 350 buses
- (F) 450 buses
- (G) 4000 buses

74. A student scores 7 out of every 20 marks on a tough test. Expressed as a percentage, the score is? [\[Comparing Quantities\]](#)

- (A) 40%
- (B) 13%
- (C) 70%
- (D) 35%
- (E) 17.5%
- (F) 27%

75. A sample of clay soil drains 150 millilitres of water in 30 minutes. At this steady rate, how long will it take to drain 250 millilitres? [\[Soil\]](#)

- (A) 45 minutes
- (B) 1250 minutes
- (C) 5 minutes
- (D) 50 minutes
- (E) 100 minutes
- (F) 60 minutes
- (G) 75 minutes

76. The expression 3 cubed plus 3 cubed plus 3 cubed can be written as a single power of 3. Which one?

[\[Exponents & Powers\]](#)

- (A) 3^5
- (B) 27^3
- (C) 9^3
- (D) 3^3
- (E) 3^9
- (F) 3^6
- (G) 3^4

77. A cyclone whose centre is 480 kilometres away is approaching at 40 kilometres per hour. How much warning time do residents have before it arrives? [\[Winds, Storms & Cyclones\]](#)

- (A) 24 hours
- (B) 15 hours
- (C) 6 hours
- (D) 19200 hours
- (E) 12 hours
- (F) 520 hours
- (G) 8 hours

78. Two lengths measure 2 metres and 50 centimetres. Written in the same unit and simplified, their ratio is? [\[Comparing Quantities\]](#)

- (A) 4 : 1
- (B) 5 : 1
- (C) 40 : 1
- (D) 2 : 50
- (E) 1 : 25
- (F) 25 : 1
- (G) 1 : 4

79. Earthworms in a plot pull about 3 grams of dead leaves underground each day. Over the 28 days of February, the total mass they bury is? [\[Soil\]](#)

- (A) 840 g
- (B) 9 g
- (C) 84 g
- (D) 112 g
- (E) 60 g
- (F) 56 g

80. Light from the Sun travels about 1.5×10^8 kilometres to reach Earth. Doubling this for a round trip and back gives a distance written as? [\[Exponents & Powers\]](#)

- (A) 1.5×10^9 km
- (B) 1.5×10^{16} km
- (C) 3×10^9 km
- (D) 3×10^{16} km
- (E) 3×10^8 km
- (F) 2.25×10^8 km
- (G) 6×10^8 km

81. Air over a desert is heated from 25 to 75 degrees Celsius. The temperature rise, as a percentage of the starting temperature, is? [\[Winds, Storms & Cyclones\]](#)

- (A) 67%
- (B) 50%
- (C) 250%
- (D) 300%
- (E) 150%
- (F) 200%
- (G) 100%

82. A family with a monthly income of Rs 12000 saves Rs 3000 of it. The percentage of income they spend is? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 30%
- (C) 75%
- (D) 80%
- (E) 40%
- (F) 70%
- (G) 60%

83. In a field, fertile topsoil makes up the top 25 centimetres of a soil profile that is 1 metre deep in total. The fraction of the profile that is topsoil is? [\[Soil\]](#)

- (A) $\frac{1}{2}$
- (B) $\frac{25}{1}$
- (C) $\frac{1}{25}$
- (D) $\frac{4}{1}$
- (E) $\frac{3}{4}$
- (F) $\frac{1}{40}$
- (G) $\frac{1}{4}$

84. Using exponent laws, the value of 2 to the fourth power multiplied by 2 to the fourth power is? [\[Exponents & Powers\]](#)

- (A) 32
- (B) 16
- (C) 128
- (D) 65536
- (E) 512
- (F) 64
- (G) 256

85. One atmosphere of pressure is about 101 kilopascals. Deep below a storm-driven wave the pressure reaches 3 atmospheres. In kilopascals, that is? [\[Winds, Storms & Cyclones\]](#)

- (A) 98 kPa
- (B) 3 kPa
- (C) 33.7 kPa
- (D) 303 kPa
- (E) 606 kPa
- (F) 104 kPa
- (G) 301 kPa

86. Rs 10000 is placed in a scheme that grows it by 10 percent each year. After 2 years its value is?

[\[Comparing Quantities\]](#)

- (A) Rs 12010
- (B) Rs 12000
- (C) Rs 12100
- (D) Rs 2100
- (E) Rs 10200
- (F) Rs 13310
- (G) Rs 11000

87. Topsoil forms very slowly, at roughly 1 centimetre every 200 years. How many years would it take for 3 centimetres of topsoil to form? [\[Soil\]](#)

- (A) 600 years
- (B) 66.7 years
- (C) 400 years
- (D) 200 years
- (E) 300 years
- (F) 603 years
- (G) 6000 years

88. Rewritten as a power of 2, the number 8 squared equals which single power, and its value is? [\[Exponents & Powers\]](#)

[\[Exponents & Powers\]](#)

- (A) $2^4 = 16$
- (B) $8^2 = 16$
- (C) $2^5 = 32$
- (D) $2^8 = 256$
- (E) $2^6 = 64$
- (F) $2^3 = 8$
- (G) $2^{16} = 65536$

89. A flag snaps in a gusting wind measured at 72 kilometres per hour. In metres per second, this gust speed is? [\[Winds, Storms & Cyclones\]](#)

- (A) 10 m/s
- (B) 2 m/s
- (C) 259 m/s
- (D) 20 m/s
- (E) 200 m/s
- (F) 25.9 m/s

90. After a 15 percent discount, a shirt sells for Rs 680. Its original marked price was? [\[Comparing Quantities\]](#)

- (A) Rs 102
- (B) Rs 720
- (C) Rs 782
- (D) Rs 765
- (E) Rs 700
- (F) Rs 800
- (G) Rs 850

91. A bucket holds 12 kilograms of moist soil that is 25 percent water by mass. The mass of dry soil in the bucket is? [\[Soil\]](#)

- (A) 9 kg
- (B) 11.75 kg
- (C) 15 kg
- (D) 48 kg
- (E) 8 kg
- (F) 6 kg

92. A single bacterium divides into 2 every hour. Starting from one cell, the number present after 6 hours, written as a power, is? [\[Exponents & Powers\]](#)

- (A) $6^2 = 36$
- (B) $2^6 = 64$
- (C) $2^3 = 8$
- (D) $2^7 = 128$
- (E) $2^5 = 32$
- (F) $12 = 12$
- (G) $2^6 = 36$

93. A cyclone's outer cloud band has a radius of 250 kilometres while its calm eye has a radius of 25 kilometres. The ratio of the band radius to the eye radius is? [\[Winds, Storms & Cyclones\]](#)

- (A) 10 : 1
- (B) 250 : 25
- (C) 1 : 10
- (D) 5 : 1
- (E) 25 : 1
- (F) 100 : 1
- (G) 2 : 1

94. A box of 50 pens costs Rs 250. If a seller wants a 20 percent profit on the whole box, the selling price of one pen should be? [\[Comparing Quantities\]](#)

- (A) Rs 300
- (B) Rs 6
- (C) Rs 5
- (D) Rs 5.50
- (E) Rs 7
- (F) Rs 6.50

95. A sloping field loses soil to erosion at 4 tonnes per hectare each year. Over 3 years, a 2 hectare field loses how much soil? [\[Soil\]](#)

- (A) 16 tonnes
- (B) 12 tonnes
- (C) 48 tonnes
- (D) 24 tonnes
- (E) 6 tonnes
- (F) 8 tonnes

96. Simplifying, 4 squared multiplied by 2 to the third power equals which power of 2, and its value?

[\[Exponents & Powers\]](#)

- (A) $2^{12} = 4096$
- (B) $8^3 = 512$
- (C) $2^7 = 128$
- (D) $2^6 = 64$
- (E) $2^7 = 64$
- (F) $2^5 = 32$

97. A weather balloon climbs at a steady 5 metres per second. How high, in kilometres, will it be after 20 minutes of rising? [\[Winds, Storms & Cyclones\]](#)

- (A) 60 km
- (B) 3 km
- (C) 6000 km
- (D) 100 km
- (E) 0.6 km
- (F) 1.2 km
- (G) 6 km

98. A cricket bat bought for Rs 1200 is sold at a loss of 15 percent. The price it fetched on sale is? [\[Comparing Quantities\]](#)

- (A) Rs 1215
- (B) Rs 180
- (C) Rs 1380
- (D) Rs 1185
- (E) Rs 1050
- (F) Rs 1000
- (G) Rs 1020

99. Freshly dug, loose soil is about 30 percent air by volume. A tin holding 2 litres of this soil contains how many millilitres of air? [\[Soil\]](#)

- (A) 2030 mL
- (B) 200 mL
- (C) 6 mL
- (D) 60 mL
- (E) 1400 mL
- (F) 600 mL
- (G) 300 mL

100. Multiplying powers of ten, 10 squared times 10 cubed equals which power of ten, and what is its value? [\[Exponents & Powers\]](#)

- (A) $10^6 = 100000$
- (B) $10^1 = 10$
- (C) $10^6 = 1000000$
- (D) $10^5 = 10000$
- (E) $10^4 = 10000$
- (F) $10^5 = 100000$
- (G) $10^5 = 1000000$